

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without permission of BIS or used as an Indian Standard)

भारतीय मानक मसौदा
एल्बी प्रकार का अस्थि शल्य चिकित्सा के लिए मेज - विशिष्टि
[IS 7596 का पहला पुनरीक्षण]

Draft Indian Standard
Albee's Type Orthopaedic Operation Table - Specification
[First Revision of IS 7596]

ICS 11.140

Hospital Equipment and Surgical Disposable
Products Sectional Committee, MHD 12

Last date for comments:
12 September 2026

FOREWORD

(Formal clause will be added later)

This standard was originally published in 1974. The first revision of this standard has been brought out to align it with recent developments in the material requirements, removal of wooden material from table top and knob and to bring the standard in line with the latest style and format of referred standards.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This standard specifies the manually controlled hydraulic Albee's type orthopaedic operation table.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

IS No.	Title
IS 6911: 2026	Stainless Steel Plate, Sheet and Strip - Specification (<i>third revision</i>)
IS 1068: 2025 ISO 1456: 2009	Metallic and other inorganic coatings - Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium (<i>fourth revision</i>)
IS 17523: 2021	Castors and Wheels - Requirements for Castors for Furniture

3 MATERIAL REQUIREMENTS

3.1 The materials used for the manufacturing of the operating table shall be corrosion-resistant or shall have a protective coating.

3.2 The raising column and the gearing system shall be made from wear-resistant material.

3.3 The knobs shall be made from ebonite or plastics that withstand mild acids.

3.4 The stainless steel used for the operation table shall conform to designation X07Cr17Ni12Mo2 in accordance with IS 6911.

3.5 The anti-static rubber tread castors shall conform to IS 17523.

4 SHAPE AND DIMENSIONS

4.1 The shape of the operation table shall be as shown in Fig. 1.

4.2 Length and Width

The overall length of the table top including the leg section shall be 1800 ± 50 mm. The overall width of the table top shall be 450 to 500 mm at the head section and 550 to 600 mm at the centre section and the leg section.

4.3 Height

The lowest position of the table top shall be 800 ± 50 mm and in the highest position 950 ± 50 mm.

5 CONSTRUCTION REQUIREMENTS

5.1 Base

5.1.1 The base shall be solid and mounted on castors to provide stability to the operation table. The design of the table shall allow manipulation of the mobile image intensifier without obstruction.

5.1.2 The base shall have the following fittings:

- The crank to raise and lower the table top;
- The crank to give the table trendelenburg motion and also actuate the compensating floor locks;
- The crank to give tilting motion to the table top.

5.2 Table Top

The table top shall have three sections such as head section, centre section and leg section. The table top shall satisfy the following requirements:

- a) The head section shall be capable of being raised or lowered independently and rotate horizontally through an angle of 360°;
- b) The centre section shall have one or two flaps which shall be removable when the patient is on the table and capable of being raised or lowered independently;
- c) The leg section shall be foldable or removable type;
- d) The entire table top shall be capable to raised or lowered and able to tilt 30° sideways;
- e) The table shall be capable of Trendelenburg movement of 30°;
- f) The table shall be provided with removable perineal posts of adult and child sizes.

5.3 Traction Apparatus

5.3.1 Lower Extremities Traction Apparatus

The operation table shall be provided with a lower extremities traction apparatus (see Fig. 1) connected through a link mechanism. The stirrups of the apparatus shall be capable of being rotated in any direction through a ball-and-socket arrangement.

5.3.2 Upper Extremities Traction Apparatus

The upper extremities traction apparatus shall consist of an overhead frame and a lower extremities traction apparatus (see 5.3.1) provided with a Weinberger traction device for gripping the fingers.

5.4 Hydraulic Lifting Mechanism

The table shall have a hydraulic pump encased in the base and operated by a crank to control the lift of the table. The table lift shall be between 150 and 200 mm. The controls shall be easily accessible to clean and maintain. The parts intended to be removed by the user shall be easy to replace.

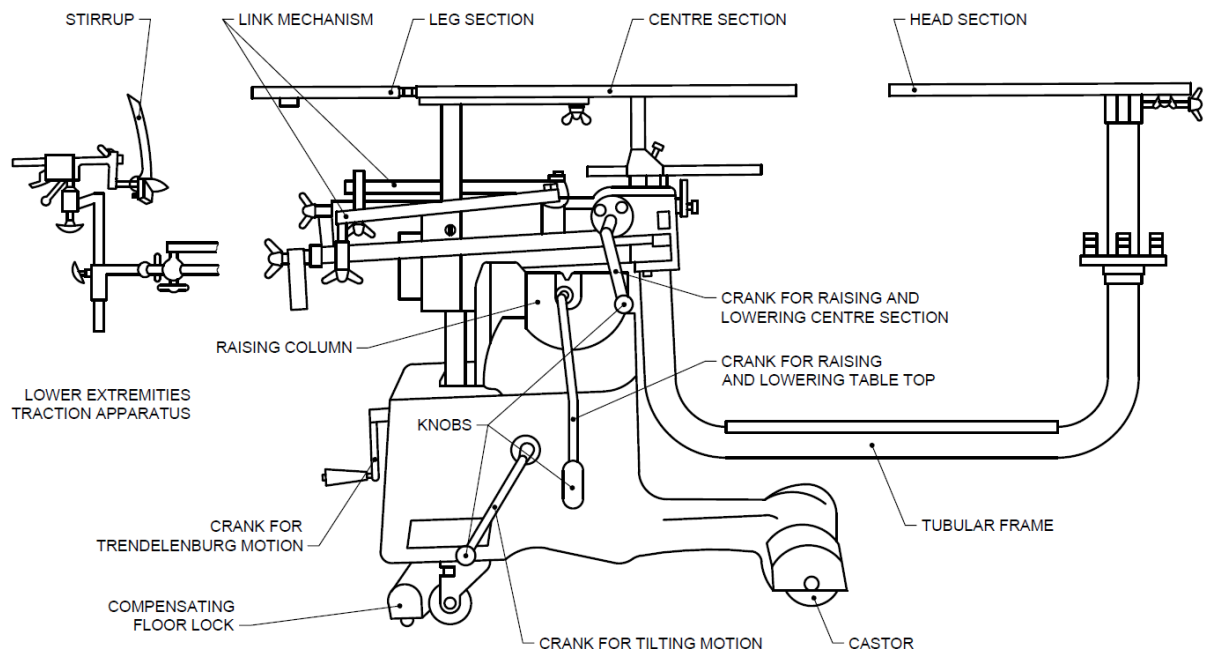


Fig. 1 ALBEE'S TYPE ORTHOPAEDIC OPERATION TABLE

6 ACCESSORIES

6.1 The following accessories shall be optional for operation table:

TABLE 1 LIST OF ACCESSORIES

Sr No.	Accessories
1	Winch and canvas with holder bracket
2	Goldthwaite irons
3	Cervical yoke and headrest
4	Intravenous arm board
5	Intramedullary nailing support
6	Cassette for lateral radiography
7	Cassette for A-P radiography
8	Footrest for hip nailing
9	Hyperextension board

7 PERFORMANCE REQUIREMENTS

7.1 The table shall be stable on castors with floor locks.

7.2 The table shall be capable of moving on castors on a smooth cemented floor from its static position when a pull of 120 N is applied and it shall be capable of propulsion with a pull of 70 N after it is set in the motion.

7.3 The floor locks shall provide rigidity to the table and the table shall not move when a force of 500 N is applied to the base in any direction.

7.4 The force of 500 N shall be applied vertically downwards on the extreme edge of the head section in any position. When the load is applied the leg section shall not lift by more than 10 mm.

7.5 The operation table shall be clamped with all accessories and checked for wobbles and movement under normal conditions.

7.6 The mass of 5 Kg shall be dropped from a height of 150 mm on the extreme end of the head section or centre section and checked for table top damage.

7.7 Hydraulic Lifting Mechanism

The hydraulic lifting mechanism shall provide smooth movement to the table top. The table top height movement from topmost position to lowest position shall be checked by releasing the hydraulic pressure, the movement time required by the table top shall be in the range of 10-30 seconds. The table top is checked by putting a mass of 250 Kg on the centre section and observed that the table shall not sink more than 3 mm in a period of 4 hours. The performance of the hydraulic lifting mechanism shall remain unimpaired when the table top with a mass of 250 kg kept on the centre section is moved up and down for 20 times.

7.8 Traction Apparatus

The traction apparatus shall be easily approachable and replaceable. The movements of the links shall be smooth.

8 WORKMANSHIP AND FINISH

8.1 The various components of the operation table shall be well finished to provide smooth movement for performance requirements laid down in 7. The surfaces shall be well finished and do not have surface defects such as pin holes, blisters and blow holes. The three sections of the table top in their normal positions shall be in one plane.

8.2 The welded joints shall have welds fully penetrating and shall be well finished.

8.3 The non-corrodible metal surfaces made with aluminium and stainless-steel shall be mat finished.

8.4 The corrodible surfaces made up of steel castings for base, cylinder and pedal shall be painted to non-reflecting paint surface such as sea green or as agreed to between the manufacturer and the purchaser. The surface which is painted shall be degreased, rustproofed by phosphating and then suitably protected by an anticorrosive primer either by brushing or by spraying and finished by spraying stoving enamel or air-drying enamel of the specified shade. The coat shall be stoved or air dried separately in each stage. The resulting finish shall be hard and shall not readily chip or flake off.

8.5 The brass and mild steel parts shall be plated with chromium over nickel and the coatings shall conform to **Service Condition 2** of IS 1068.

8.6 The knobs shall provide an easy grip.

9 OPERATING MANUAL

The instruction manual giving following information shall be supplied with the operation table:

- a) Brief instructions for installation and maintenance;
- b) Lift of the hydraulic lifting mechanism;
- c) Instructions for correct operation of the table.

10 MARKING

10.1 The marking which is used by the manufacturer shall be legibly and indelibly marked with the Identification mark of the manufacturer.

10.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

11 PACKAGING

The operation table shall be packed as agreed to between the purchaser and the manufacturer. Each individual package shall contain only one unit and may be marked with the name of the item, and identification of the source of manufacture and batch number and any other marking if so desired.